Hoang Trieu Vy Le

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Summary — I am a Senior Research Scientist at Beink Dream, specializing in 3D lightweight learning and optimization. Previously at CEA (France) and PhD from École Normale Supérieure under Prof. Nelly Pustelnik. My work bridges physics, AI, and efficient model design for 2D/3D applications — with a strong drive toward physics research: learning compact, physically grounded representations from continuous high-dimensional signals.

Experiences

Beink Dream **Mar 2025 – Present**
Senior Research Scientist *Paris, France*
Architected end-to-end agentic AI systems integrating LLM reasoning, generative vision models, and structured 3D/CAD output — building multi-modal systems that maintain a structured internal representation of scene state across 2D, 3D, and CAD modalities.

CEA – DIGITEO Labs **Feb 2024 – Feb 2025**
Research Engineer *Saclay, France*
Worked on lightweight learning models for 3D CT reconstruction — physics-consistent scene representations from sparse, noisy measurements and limited high-dimensional observations.

INMA, ICTEAM – UC Louvain **Sep 2020 – Aug 2021**
Visiting PhD Researcher *Louvain, Belgium*
Designed scalable accelerated algorithms to solve joint image restoration and edge detection on large-scale physics images.

OGoxe – Météo France **Jan 2020 – Aug 2020**
Research Intern *Toulouse, France*
Built velocity estimation models from video sequences — learning to infer physical dynamics (fluid flow) from continuous visual observations, an early instance of physics-grounded world dynamics modeling.

IMFT – Toulouse Fluid Mechanics Institute **Jul 2019 – Oct 2019**
Research Intern – High Performance Computing *Toulouse, France*
Worked on a novel Bucketing-based algorithm for particle-based simulation using parallelized mesh-less models across multi-GPU clusters.

Education

École Normale Supérieure de Lyon **Sep 2020 – Dec 2023**
Ph.D. in Applied Mathematics, AI, Signal and Image Processing *Lyon, France*
Thesis: *Variations on the Mumford-Shah functional for interface detection in degraded images: from proximal algorithms to unrolled architectures.*
b Advised by Prof. Nelly Pustelnik & Dr. Marion Foare.

INSA Toulouse **Sep 2015 – Sep 2020**
Master of Engineering in Applied Mathematics *Toulouse, France*
Major in Mathematical Modelling and Numerical Analysis.

Publications

Unfolded Discrete Mumford-Shah Functional for Joint Image Denoising and Edge Detection, *EUSIPCO 2025*
H.T.V Le, Marion Foare, Audrey Repetti, Nelly Pustelnik

Proximal Neural Networks based Reconstruction for Few-View CT Applications, *ICT 2025*
H.T.V Le, C. Bossuyt, J. Escoda, H. Wang, J. De Beenhouwer, J. Sijbers

Embedding Blake-Zisserman Regularization in Unfolded PNNs for Enhanced Edge Detection, *Under review*
H.T.V Le, Marion Foare, Audrey Repetti, Nelly Pustelnik

Unfolded Proximal Neural Networks for Robust Image Gaussian Denoising, *IEEE TIP 2024*
H.T.V Le, Audrey Repetti, Nelly Pustelnik

Proximal Based Strategies for Solving Discrete Mumford-Shah With Ambrosio-Tortorelli Penalization on Edges, *IEEE SPL 2022*
H.T.V Le, Marion Foare, Nelly Pustelnik

The Faster Proximal Algorithm, the Better Unfolded Deep Learning Architecture? A Study Case of Image Denoising, *EUSIPCO 2022*
H.T.V Le, Nelly Pustelnik, Marion Foare

Fast Unfolded Proximal Algorithms for the Analysis of Piecewise Homogeneous Fractal Images, *Gretsi 2022*
H.T.V Le, Barbara Pascal, Nelly Pustelnik, Marion Foare, Patrice Abry

Activities

Teaching: Signal processing, convex optimization, unfolded NNs and inverse problems (ENS Lyon); Signal processing and Cryptography Mathematics (EPITA Lyon); Fundamental Machine Learning (EPITA Paris)

Talks: EUSIPCO, Belgrade, Serbia **Aug 2022**
Conference on Digital Geometry and Discrete Variational Calculus **Mar 2021**

Open-Source Software:

P-DMS-AT [Python] - Proximal strategies for Discrete Mumford-Shah with Ambrosio-Tortorelli edge penalization.
Proximal Neural Networks [PyTorch] — Lightweight unrolled architectures combining iterative algorithms and deep networks for inverse problems.
BZ-PNN [PyTorch] — Super-lightweight model for enhanced edge detection via Blake-Zisserman regularization.
DMS-PNN [PyTorch] — Unfolded Discrete-Mumford-Shah Proximal Neural Networks.
Bucketing-based Algorithm [Fortran, C/C++, Python] — Parallelized linearly transformed particle method for diffusion equations.

Awards

- Third prize, Vietnam National Physics Olympiad **2015**
- Fourth prize, Vietnam National Physics Olympiad **2014**
- Silver medal, Northern Regional Vietnam Physics Olympiad **2013**